BAEKSEOK UNIVERSITY

chap 05. C Operators

Contents

01 블랙박스 테스트 개요

Operators

- C divides the operators into the following groups:
  - Arithmetic operators
  - Assignment operators
  - Comparison operators
  - Logical operators
  - Bitwise operators (advanced)

Arithmetic Operators

Operator	Name	Example
+	Addition	x + y
-	Subtraction	x - y
*	Multiplication	x * y
/	Division	x / y
%	Modulus	x % y
++	Increment	++x
--	Decrement	--x

```c
int x = 10;
int y = 3;

printf("%d\n", x + y); // 13
printf("%d\n", x - y); // 7
printf("%d\n", x * y); // 30
printf("%d\n", x / y); // 3
printf("%d\n", x % y); // 1

int z = 5;
++z;
printf("%d\n", z); // 6
--z;
printf("%d\n", z); // 5

int a = 10;
int b = 3;
printf("%d\n", a / b);   // Integer division, result is 3

double c = 10.0;
double d = 3.0;
printf("%f\n", c / d);   // Decimal division, result is 3.333...
```

Arithmetic Operators

- Incrementing and Decrementing
  - Incrementing and decrementing are very common in programming, especially when working with counters, loops, and arrays

```c
int x = 5;

++x; // Increment x by 1
printf("%d\n", x); // 6

int x = 5;

--x; // Decrement x by 1
printf("%d\n", x); // 4
```

```c
int peopleInRoom = 0;

// 3 people enter
peopleInRoom++;
peopleInRoom++;
peopleInRoom++;

printf("%d\n", peopleInRoom); // 3

// 1 person leaves
peopleInRoom--;

printf("%d\n", peopleInRoom); // 2
```

Assignment

Operator	Example	Same As
=	x = 5	x = 5
+=	x += 3	x = x + 3
-=	x -= 3	x = x - 3
*=	x *= 3	x = x * 3
/=	x /= 3	x = x / 3
%=	x %= 3	x = x % 3
&=	x &= 3	x = x & 3
|=	x |= 3	x = x | 3
^=	x ^= 3	x = x ^ 3
>>=	x >>= 3	x = x >> 3
<<=	x <<= 3	x = x << 3

Comparison Operators

Comparison Operators

- Comparison operators are used to compare two values (or variables)
- The return value of a comparison is either 1 or 0, which means true (1) or false (0)

perator	Name	Example	Description
==	Equal to	x == y	Returns 1 if the values are equal
!=	Not equal	x != y	Returns 1 if the values are not equal
>	Greater than	x > y	Returns 1 if the first value is greater t han the second value
<	Less than	x < y	Returns 1 if the first value is less than the second value
>=	Greater than or equa l to	x >= y	Returns 1 if the first value is greater t han, or equal to, the second value
<=	Less than or equal to	x <= y	Returns 1 if the first value is less than , or equal to, the second value

Comparison Operators

- Examples

```
1 int age = 18;
2
3 printf("%d\n", age >= 18); // 1 (true), old enough to vote
4 printf("%d\n", age < 18);  // 0 (false)
5
6 nt passwordLength = 5;
7
8 printf("%d\n", passwordLength >= 8); // 0 (false), too short
9 printf("%d\n", passwordLength < 8);  // 1 (true), needs more characters
```

Logical Operators

Logical Operators

- Logical operators are used to determine the logic between variables or values, by combining multiple conditions:

Operator	Name	Example	Description
&&	AND	x < 5 && x < 10	Returns 1 if both statements are true
||	OR	x < 5 || x < 4	Returns 1 if one of the statements is true
!	NOT	!(x < 5 && x < 10)	Reverse the result, returns 0 if the result is 1

```c
int isLoggedIn = 1;
int isAdmin = 0;

printf("Regular user: %d\n", isLoggedIn && !isAdmin);
printf("Has access: %d\n", isLoggedIn || isAdmin);
printf("Not logged in: %d\n", !isLoggedIn);
```

Operator Precedence

Operator Precedence

- When a calculation contains more than one operator, C follows **order of operations** rules to decide which part to calculate first.

Order of Operations

- () - Parentheses
- *, /, % - Multiplication, Division, Modulus
- +, - - Addition, Subtraction
- >, <, >=, <= - Comparison
- ==, != - Equality
- && - Logical AND
- || - Logical OR
- = - Assignment

Challenge: Calculate the Total Cost of an Item

- Instructions
  - Inside main(), complete the following steps:
    1. Declare two int variables named itemPrice(50) and shippingCost(15), and assign them values
    2. Create an int variable named sum
    3. Calculate the total cost by adding itemPrice and shippingCost (store the result in sum)
    4. Print the total cost using printf

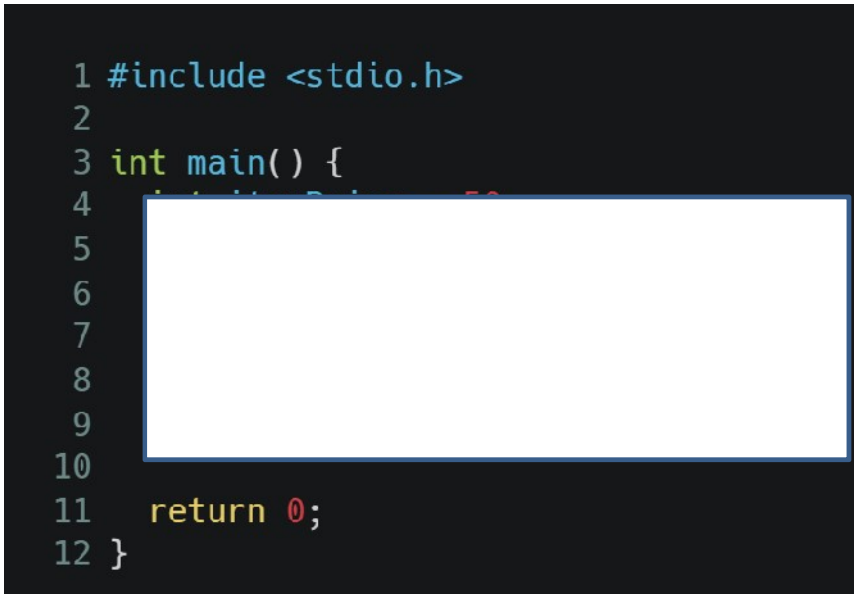
```c
#include <stdio.h>

int main() {
  // Write itemPrice here
  // Write shippingCost here

  // Write sum here (itemPrice + shippingCost)

  // Print sum here

  return 0;
}
```



User Input

- user Input
  - To get user input, you can use the scanf() function:

```c
// Create an integer variable that will
// store the number we get from the user
int myNum;

// Ask the user to type a number
printf("Type a number: \n");

// Get and save the number the user types
scanf("%d", &myNum);

// Output the number the user typed
printf("Your number is: %d", myNum);
```

```c
// Create an int and a char variable
int myNum;
char myChar;

// Ask the user to type a number AND a character
printf("Type a number AND a character and press enter: \n");

// Get and save the number AND character the user types
scanf("%d %c", &myNum, &myChar);

// Print the number
printf("Your number is: %d\n", myNum);

// Print the character
printf("Your character is: %c\n", myChar);
```

Code challenge

- Exchange rate calculation
- Instruction
  - Inside main(), complete the following steps:
    1. Declare two int variables named nDollar , nWon
    2. print input current Won per dollar
    3. get from user nDollar
    4. print input total Won
    5. get from user nWon
    6. Calculate and print exchanged dollar

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Thank You !